

```
from flask import Flask, request, session, redirect, render_template_string
```

```
import sqlite3
```

```
import bcrypt
```

```
app = Flask(__name__)
```

```
app.secret_key = "my_secret_key"
```

```
# Create database
```

```
def init_db():
```

```
    conn = sqlite3.connect("users.db")
```

```
    cursor = conn.cursor()
```

```
    cursor.execute("""
```

```
        CREATE TABLE IF NOT EXISTS users (
```

```
            id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
            username TEXT UNIQUE NOT NULL,
```

```
            password TEXT NOT NULL
```

```
        )
```

```
    """)
```

```
    conn.commit()
```

```
    conn.close()
```

```
init_db()
```

```
# Home page
```

```
@app.route("/")
```

```
def home():
```

```
    if "username" in session:
```

```
        return f"""
```

```
<h1>Welcome, {session['username']}!</h1>
```

```
<a href="/logout">Logout</a>
```

```
"""
```

```
return ""
```

```
<h1>Secure Login System</h1>
```

```
<a href="/register">Register</a><br><br>
```

```
<a href="/login">Login</a>
```

```
"""
```

```
# Register
```

```
@app.route("/register", methods=["GET", "POST"])
```

```
def register():
```

```
    if request.method == "POST":
```

```
        username = request.form["username"]
```

```
        password = request.form["password"]
```

```
        if len(username) < 3 or len(password) < 6:
```

```
            return "Username must have 3 characters and password must have 6 characters."
```

```
        hashed_password = bcrypt.hashpw(
```

```
            password.encode("utf-8"),
```

```
            bcrypt.gensalt()
```

```
        )
```

```
    try:
```

```
        conn = sqlite3.connect("users.db")
```

```
        cursor = conn.cursor()
```

```

        cursor.execute(
            "INSERT INTO users (username, password) VALUES (?, ?)",
            (username, hashed_password)
        )

        conn.commit()

        conn.close()

    return redirect("/login")

except sqlite3.IntegrityError:
    return "Username already exists!"

return ""

<h2>Register</h2>

<form method="POST">
    Username:<br>
    <input type="text" name="username" required><br><br>

    Password:<br>
    <input type="password" name="password" required><br><br>

    <button type="submit">Register</button>
</form>

<br>

<a href="/">Home</a>

""

```

```
# Login

@app.route("/login", methods=["GET", "POST"])
def login():

    if request.method == "POST":

        username = request.form["username"]
        password = request.form["password"]

        conn = sqlite3.connect("users.db")
        cursor = conn.cursor()

        cursor.execute(
            "SELECT password FROM users WHERE username = ?",
            (username,)
        )

        user = cursor.fetchone()
        conn.close()

        if user and bcrypt.checkpw(
            password.encode("utf-8"),
            user[0]
        ):
            session["username"] = username
            return redirect("/")

        return "Invalid username or password!"

    return ""
```

```
<h2>Login</h2>
```

```
<form method="POST">
```

```
    Username:<br>
```

```
    <input type="text" name="username" required><br><br>
```

```
    Password:<br>
```

```
    <input type="password" name="password" required><br><br>
```

```
    <button type="submit">Login</button>
```

```
</form>
```

```
<br>
```

```
<a href="/">Home</a>
```

```
"""
```

```
# Logout
```

```
@app.route("/logout")
```

```
def logout():
```

```
    session.pop("username", None)
```

```
    return redirect("/")
```

```
if __name__ == "__main__":
```

```
    app.run(debug=True)
```

Register

Username:

Password:

Register

Login

Username:

Password:

Login

[Home](#)

[Home](#)

Welcome, 123456!

[Logout](#)

Secure Login System

[Register](#)

[Login](#)